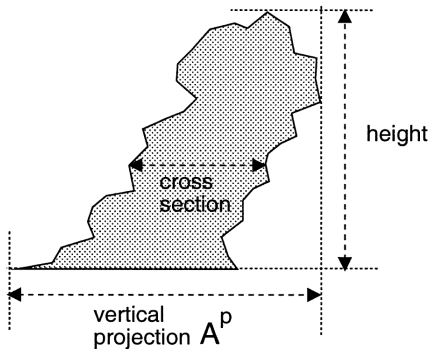


Large-eddy simulations (LES) of shallow cumulus: simulated clouds, not measurements. Each cloud gets one size, and the paper studies, among other things, the histogram of these sizes.



$$l = \sqrt{A^p}$$

$$N = \int_0^\infty \mathcal{N}(l) dl$$

Three simulated cases, set up like real campaigns:

- BOMEX: Barbados, ocean.
- SCMS: Florida, land.
- ARM: Oklahoma, land, diurnal cycle.

Symbols

cloud: group of touching saturated grid cells in 3D

A^p vertically projected area of one cloud: its footprint seen from above

l cloud size, m

$\mathcal{N}(l)$ number of clouds per unit size, m^{-1}

N total number of clouds in the domain

Find each cloud in 3D $\rightarrow$ project it vertically $\rightarrow$ histogram of the sizes l

A power law below a variable size break

2

The histogram counts clouds per size: **small clouds are by far the most numerous**. Below a size ℓ_d their number falls as a power law of size; above ℓ_d it falls faster.

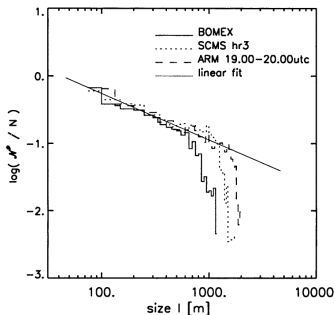


Fig. 4a: size density against l .

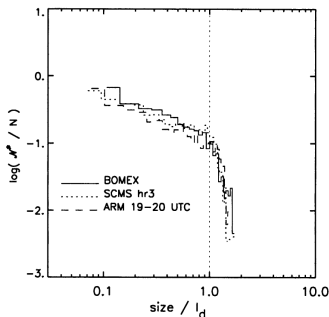


Fig. 6: against l/ℓ_d , collapsed.

$$\mathcal{N}(l) = al^b$$

$$b = -1.70 \text{ for } l < \ell_d$$

Symbols

ℓ_d scale break: size where the exponent changes. BOMEX 700 m, SCMS 1050 m, ARM 400 $\rightarrow$ 1250 m in the afternoon

a, b fit constants; b is the exponent
 $\mathcal{N}^* = l \ln(10) \mathcal{N}$, clouds per decade of size (plotted), slope $b + 1 = -0.70$

Same exponent in all cases: only the break ℓ_d changes

The size break sets the dominant cloud cover

3

Weighting every cloud by its area gives the cloud cover contributed by each size. It grows up to ℓ_d and drops beyond, so **mid-size clouds cover most of the sky**.

$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y} \propto l^{b+2} = l^{0.30} \quad (l < \ell_d)$$

$$a^p = \int_0^\infty \alpha^p(l) dl$$

Symbols

$\alpha^p(l)$ cloud cover from clouds of size l , per unit size, m^{-1}

a^p projected cloud cover: fraction of the domain covered, seen from above

$L_x L_y$ horizontal area of the domain

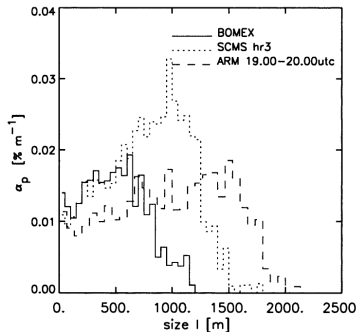


Fig. 7a: $\alpha^p(l)$, three cases.

Cloud cover peaks near the break ℓ_d

Mass flux also peaks at intermediate sizes

4

Larger clouds rise faster, so mass flux still peaks at **intermediate** sizes, near the cover peak, but the smallest clouds drop out: their vertical velocity is very low.

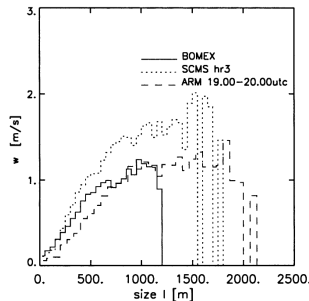


Fig. 7b: mean vertical velocity w .

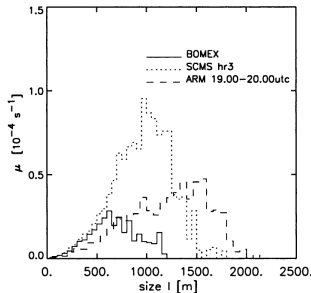


Fig. 7c: mass flux $\mu(l)$.

Mass flux:

$$\mu(l, z) = \alpha(l, z) w(l, z)$$

$$\mu(l) = \frac{1}{h_c} \int_{h_c} \mu(l, z) dz$$

Symbols

$w(l, z)$ mean vertical velocity of clouds of size l at height z , m s^{-1}

$\alpha(l, z)$ their cloud cover at height z , per unit size

h_c depth of the cloud layer

μ mass flux divided by air density, per unit size, s^{-1}

Same intermediate dominant size as cover; the smallest clouds carry almost nothing

“Climbable clouds” are far wider than thermals

5

The paper gives no thermal statistics. Fig. 7b gives the mean updraft w of each cloud size: keep the sizes where w beats a pilot's sink rate. Only the two land cases, SCMS and ARM, are used.

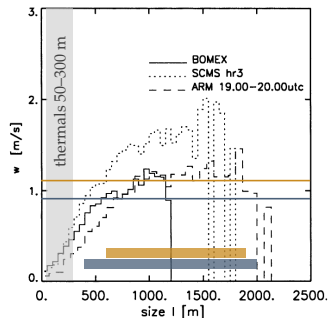


Fig. 7b; overlay ours.

Climbing needs w above the sink rate

w : mean updraft of clouds of size l . Sink rates are for straight flight: a circling wing sinks faster, so **the real min. sink thresholds are higher**.

	min. sink	clouds with $w \geq$ sink	thermal diameter
Paraglider	1.1 m/s	0.6–1.9 km	50–300 m
Hang glider	0.9 m/s	0.4–2.0 km	80–300 m

Clouds of 50–300 m are the most numerous, but their mean updraft stays below 0.7 m/s: too weak to keep a paraglider or a hang glider climbing.

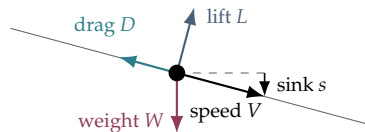
Clouds with enough updraft are 0.4–2 km wide: several times, up to ten times, the 50–300 m expected for the thermals pilots use.

“Climbable clouds”: 0.4–2 km. Expected thermals: 50–300 m

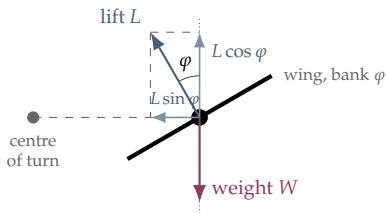
Appendix: sink rate in a banked turn

6

Why circling sinks faster than straight flight, at the same angle of attack.



side view, straight glide



back view, turn

1 Energy. Lift does no work; drag uses $D V$ per second, paid for by lost height

$$W s = D V$$

2 Glide ratio $E = L/D$ is set by the angle of attack, so $D = L/E$

$$s = V L / (E W)$$

3 Straight flight: lift $\approx$ weight, as the path is only $\approx 6^\circ$ below horizontal

$$L \approx W, \quad s_0 \approx V_0 / E$$

4 Turn: the vertical part of lift holds the weight, $L \cos \varphi = W$

$$L = W / \cos \varphi$$

5 Same angle of attack: lift grows as V^2

$$V = V_0 / \sqrt{\cos \varphi}$$

Put 3, 4, 5 into 2:

$$s = \frac{V_0}{\sqrt{\cos \varphi}} \cdot \frac{W}{\cos \varphi} \cdot \frac{1}{E W} = \frac{s_0}{(\cos \varphi)^{3/2}}$$

Circling sinks faster: $\times 1.10$ at 20° of bank, $\times 1.24$ at 30° , $\times 1.68$ at 45°

Steady flight; small glide angle, so $L \approx W$ (error below 1% for a glide ratio of 9). The extra drag of turning is neglected, so the true factor is larger.